MTH129 Introduction to AI for Mathematical Sciences (MS)

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SMP AI Team

Xi'an Jiaotong-Liverpool University
西交利物浦大学

Weighted Least Squares

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Introduction to Weighted Least Squares

Core idea: Modify ordinary least squares to handle heteroscedastic data

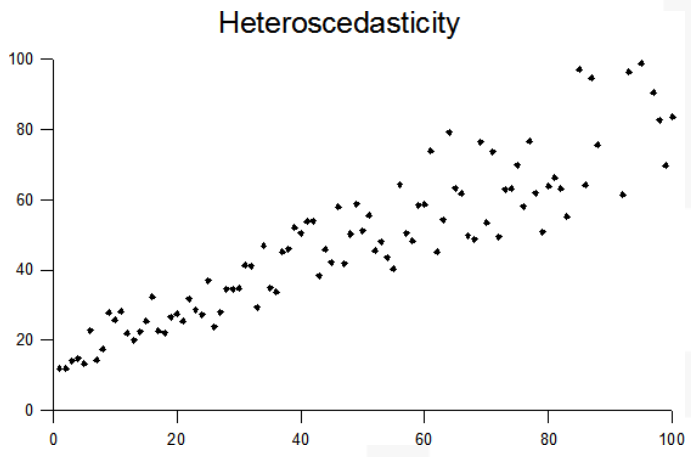
The Heteroscedasticity Problem

- Standard OLS assumes $\text{Var}(\varepsilon_i) = \sigma^2$ (constant variance)
- Real-world data often violates this: $\text{Var}(\varepsilon_i) = \sigma_i^2$
  - Measurement precision varies across observations
  - Variance depends on predictors (e.g., wealth vs. spending)
  - Data aggregated from different sources

Introduction to Weighted Least Squares

Consequences of ignoring heteroscedasticity:

- OLS estimates remain unbiased
- But not efficient (higher variance than necessary)
- Standard errors incorrect → misleading hypothesis tests
- Confidence intervals wrong coverage



WLS Solution

Assign weights proportional to observation precision:

$$d_i = \frac{1}{\sigma_i^2}, \quad \mathbf{D} = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix}$$

Weighted Least Squares (WLS) for heteroscedastic data:

Mathematical Formulation

$$\min_{\mathbf{w}} (\mathbf{y} - \mathbf{X}\mathbf{w})^T \mathbf{D} (\mathbf{y} - \mathbf{X}\mathbf{w})$$

where $\mathbf{D} = \mathbf{\Sigma}^{-1}$ is the weight matrix (inverse covariance).

- $\mathbf{D}$ diagonal for uncorrelated errors
- $\mathbf{D}$ full matrix for correlated errors (GLS)

Important

Standard least squares is special case where $\mathbf{D} = \frac{1}{\sigma^2}\mathbf{I}$

WLS Solution

Weight observations by inverse variance:

$$d_i = \frac{1}{\sigma_i^2}, \quad \mathbf{D} = \mathrm{diag}(d_1, \ldots, d_n) = \mathbf{\Sigma}^{-1}$$

WLS Objective Function

Weighted Residual Sum of Squares:

$$J(\mathbf{w}) = (\mathbf{y} - \mathbf{X}\mathbf{w})^T \mathbf{D} (\mathbf{y} - \mathbf{X}\mathbf{w})$$

where $\mathbf{D} = \mathbf{\Sigma}^{-1}$ is the weight matrix (diagonal for uncorrelated errors)

Expanding the quadratic form:

$$\begin{aligned} J(\mathbf{w}) &= \mathbf{y}^T\mathbf{D}\mathbf{y} - \mathbf{y}^T\mathbf{D}\mathbf{X}\mathbf{w} - \mathbf{w}^T\mathbf{X}^T\mathbf{D}\mathbf{y} + \mathbf{w}^T\mathbf{X}^T\mathbf{D}\mathbf{X}\mathbf{w} \\ &= \mathbf{y}^T\mathbf{D}\mathbf{y} - 2\mathbf{w}^T\mathbf{X}^T\mathbf{D}\mathbf{y} + \mathbf{w}^T\mathbf{X}^T\mathbf{D}\mathbf{X}\mathbf{w} \end{aligned}$$

- $\mathbf{D}$ symmetric $\rightarrow \mathbf{y}^T\mathbf{D}\mathbf{X}\mathbf{w} = (\mathbf{X}^T\mathbf{D}\mathbf{y})^T\mathbf{w}$
- $\mathbf{X}^T\mathbf{D}\mathbf{X}$ symmetric positive semi-definite

Gradient and Closed-form Solution

Take gradient w.r.t w:

$$\begin{aligned}\nabla_{\mathbf{w}} J(\mathbf{w}) &= \nabla_{\mathbf{w}}(\mathbf{y}^T\mathbf{D}\mathbf{y}) - 2\nabla_{\mathbf{w}}(\mathbf{w}^T\mathbf{X}^T\mathbf{D}\mathbf{y}) + \nabla_{\mathbf{w}}(\mathbf{w}^T\mathbf{X}^T\mathbf{D}\mathbf{X}\mathbf{w}) \\ &= 0 - 2\mathbf{X}^T\mathbf{D}\mathbf{y} + 2\mathbf{X}^T\mathbf{D}\mathbf{X}\mathbf{w}\end{aligned}$$

Set gradient to zero:

$$\begin{aligned}-2\mathbf{X}^T\mathbf{D}\mathbf{y} + 2\mathbf{X}^T\mathbf{D}\mathbf{X}\mathbf{w} &= 0 \\ \mathbf{X}^T\mathbf{D}\mathbf{X}\mathbf{w} &= \mathbf{X}^T\mathbf{D}\mathbf{y}\end{aligned}$$

Closed-form solution:

$$\hat{\mathbf{w}}_{\text{WLS}} = (\mathbf{X}^T\mathbf{D}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{D}\mathbf{y}$$

Special case

When $\mathbf{D} = \frac{1}{\sigma^2}\mathbf{I}$, reduces to OLS:

$$\hat{\mathbf{w}}_{\text{OLS}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Statistical Perspective for WLS

Distributional assumption:

$$\mathbf{y}|\mathbf{X}, \mathbf{w} \sim \mathcal{N}(\mathbf{X}\mathbf{w}, \mathbf{\Sigma})$$

where $\mathbf{\Sigma} = \text{diag}(\sigma_1^2, \ldots, \sigma_n^2)$ (diagonal covariance)
Probability density function:

$$p(\mathbf{y}|\mathbf{X}, \mathbf{w}) = \frac{1}{(2\pi)^{n/2}|\mathbf{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{y} - \mathbf{X}\mathbf{w})^T \mathbf{\Sigma}^{-1}(\mathbf{y} - \mathbf{X}\mathbf{w})\right)$$

Log-likelihood function:

$$\begin{aligned}\ell(\mathbf{w}) &= \log p(\mathbf{y}|\mathbf{X}, \mathbf{w}) \\ &= -\frac{n}{2}\log(2\pi) - \frac{1}{2}\log|\mathbf{\Sigma}| - \frac{1}{2}(\mathbf{y} - \mathbf{X}\mathbf{w})^T \mathbf{\Sigma}^{-1}(\mathbf{y} - \mathbf{X}\mathbf{w})\end{aligned}$$

MLE Derivation

Maximize log-likelihood:

$$\hat{\mathbf{w}}_{\text{MLE}} = \arg\max_{\mathbf{w}} \ell(\mathbf{w})$$

Since $-\frac{n}{2}\log(2\pi)$ and $-\frac{1}{2}\log|\mathbf{\Sigma}|$ are constant w.r.t $\mathbf{w}$:

$$\arg\max_{\mathbf{w}} \ell(\mathbf{w}) = \arg\min_{\mathbf{w}} (\mathbf{y} - \mathbf{X}\mathbf{w})^T \mathbf{\Sigma}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{w})$$

Recognize:

$$\mathbf{\Sigma}^{-1} = \mathbf{D} \quad \text{(weight matrix)}$$

So:

$$\hat{\mathbf{w}}_{\text{MLE}} = \arg\min_{\mathbf{w}} (\mathbf{y} - \mathbf{X}\mathbf{w})^T \mathbf{D} (\mathbf{y} - \mathbf{X}\mathbf{w})$$

Solution:

$$\hat{\mathbf{w}}_{\text{MLE}} = (\mathbf{X}^T\mathbf{D}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{D}\mathbf{y}$$

Same as direct least squares solution!

Connection and Interpretation

Theorem

Under Gaussian error assumption with covariance $\mathbf{\Sigma}$:

$$\textit{WLS estimator} = \textit{MLE}$$

Statistical properties: (Other modules will explain!)

- Unbiased: $E[\hat{\mathbf{w}}] = \mathbf{w}$
- Minimum variance: $\mathrm{Var}(\hat{\mathbf{w}}) = (\mathbf{X}^T\mathbf{D}\mathbf{X})^{-1}$
- Asymptotically normal

Practical Implementation

How to determine weights? Usually unknown in practice

1. **Two-step procedure:**
   1. Fit OLS model, get residuals $\hat{\varepsilon}_i$
   2. Estimate variances: $\hat{\sigma}_i^2 = \hat{\varepsilon}_i^2$ or smoothed version
   3. Set $\hat{d}_i = 1/\hat{\sigma}_i^2$
2. **Variance function modeling:**

$$\sigma_i^2 = \exp(\mathbf{z}_i^T \gamma) \quad \text{(log-linear model)}$$

   where $\mathbf{z}_i$ are variance predictors
3. **Iterative reweighting:**
   - Initialize weights $\hat{d}_i^{(0)} = 1$
   - For $t = 1, 2, \ldots$ until convergence:

$$\hat{\mathbf{w}} = (\mathbf{X}^T \hat{\mathbf{D}}^{(t-1)} \mathbf{X})^{-1} \mathbf{X}^T \hat{\mathbf{D}}^{(t-1)} \mathbf{y}$$
$$\hat{\sigma}_i^{2(t)} = f(\hat{\varepsilon}_i^{(t)}) \quad \text{(variance estimator)}$$
$$\hat{d}_i^{(t)} = 1/\hat{\sigma}_i^{2(t)}$$

Example code (Python)

```
import statsmodels.api as sm
mod_wls = sm.WLS(y, X, weights=1/var_est)
res_wls = mod_wls.fit()
```

WLS Iterative Example: Income vs. Consumption

Problem: Model consumption (y) vs. income (x) with heteroscedastic errors

Income (x_i)	Consumption (y_i)	True σ_i
10	12	1.0
20	22	1.5
30	27	2.0
40	38	2.5

Model:

$$y = w_0 + w_1 x + \varepsilon, \quad \mathrm{Var}(\varepsilon_i) \propto x_i$$

Iteration 0: OLS initialization

Iteration 0: OLS initialization

$$\mathbf{X} = \begin{bmatrix} 1 & 10 \\ 1 & 20 \\ 1 & 30 \\ 1 & 40 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 12 \\ 22 \\ 27 \\ 38 \end{bmatrix}$$

$$\hat{\mathbf{w}}_{\text{OLS}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 4 & 100 \\ 100 & 3000 \end{bmatrix}, (\mathbf{X}^T\mathbf{X})^{-1} = \frac{1}{2000}\begin{bmatrix} 3000 & -100 \\ -100 & 4 \end{bmatrix}$$

$$\mathbf{X}^T\mathbf{y} = \begin{bmatrix} 99 \\ 2830 \end{bmatrix}$$

$$\hat{\mathbf{w}}^{(0)} = \begin{bmatrix} 8.5 \\ 0.75 \end{bmatrix} \quad (\text{Consumption} = 8.5 + 0.75 \times \text{Income})$$

Iteration 1: Estimate Variances

Step 1: Compute residuals

$$\hat{y}_1^{(0)} = 8.5 + 0.75 \times 10 = 16.0, \qquad \hat{\varepsilon}_1 = 12 - 16.0 = -4.0$$
$$\hat{y}_2^{(0)} = 8.5 + 0.75 \times 20 = 23.5, \qquad \hat{\varepsilon}_2 = 22 - 23.5 = -1.5$$
$$\hat{y}_3^{(0)} = 8.5 + 0.75 \times 30 = 31.0, \qquad \hat{\varepsilon}_3 = 27 - 31.0 = -4.0$$
$$\hat{y}_4^{(0)} = 8.5 + 0.75 \times 40 = 38.5, \qquad \hat{\varepsilon}_4 = 38 - 38.5 = -0.5$$

Step 2: Estimate variances Assume $\text{Var}(\varepsilon_i) = kx_i$ (proportional to income)
Estimate k using residuals:

$$\begin{aligned}\hat{k} &= \frac{1}{n}\sum \frac{\hat{\varepsilon}_i^2}{x_i} \\ &= \frac{1}{4}\left(\frac{16}{10} + \frac{2.25}{20} + \frac{16}{30} + \frac{0.25}{40}\right) \\ &= \frac{1}{4}(1.6 + 0.1125 + 0.5333 + 0.00625) \\ &= 0.563\end{aligned}$$

Step 3: Compute weights

$$d_i = 1/\hat{\sigma}_i^2 = 1/(\hat{k}x_i)$$

$$\implies \mathbf{D}^{(1)} = \text{diag}\left(\frac{1}{0.563 \times 10}, \frac{1}{0.563 \times 20}, \frac{1}{0.563 \times 30}, \frac{1}{0.563 \times 40}\right)$$

$$= \begin{bmatrix} 0.177 & 0 & 0 & 0 & \\ 0 & 0.089 & 0 & 0 & \\ 0 & 0 & 0.059 & 0 & \\ 0 & 0 & 0 & 0 & 0.044 \end{bmatrix}$$

Step 4: Compute WLS estimate

$$\mathbf{X}^T\mathbf{D}^{(1)}\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 10 & 20 & 30 & 40 \end{bmatrix} \begin{bmatrix} 0.177 & 0 & 0 & 0 \\ 0 & 0.089 & 0 & 0 \\ 0 & 0 & 0.059 & 0 \\ 0 & 0 & 0 & 0.044 \end{bmatrix} \begin{bmatrix} 1 & 10 \\ 1 & 20 \\ 1 & 30 \\ 1 & 40 \end{bmatrix}$$

$$= \begin{bmatrix} 0.177 & 0.089 & 0.059 & 0.044 \\ 1.77 & 1.78 & 1.77 & 1.76 \end{bmatrix} \begin{bmatrix} 1 & 10 \\ 1 & 20 \\ 1 & 30 \\ 1 & 40 \end{bmatrix}$$

$$= \begin{bmatrix} 0.369 & 5.89 \\ 5.89 & 130.7 \end{bmatrix}$$

$$\mathbf{X}^T\mathbf{D}^{(1)}\mathbf{y} = \begin{bmatrix} 0.177 \times 12 + 0.089 \times 22 + 0.059 \times 27 + 0.044 \times 38 \\ 0.177 \times 120 + 0.089 \times 440 + 0.059 \times 810 + 0.044 \times 1520 \end{bmatrix}$$

$$= \begin{bmatrix} 6.055 \\ 108.3 \end{bmatrix}$$

$$\hat{\mathbf{w}}^{(1)} = \left(\begin{bmatrix} 0.369 & 5.89 \\ 5.89 & 130.7 \end{bmatrix} \right)^{-1} \begin{bmatrix} 6.055 \\ 108.3 \end{bmatrix} = \begin{bmatrix} 10.2 \\ 0.70 \end{bmatrix}$$

Iteration 2: Refine Weights

Step 1: New residuals

$$\hat{y}_1^{(1)} = 10.2 + 0.70 \times 10 = 17.2, \qquad \hat{\varepsilon}_1 = 12 - 17.2 = -5.2$$
$$\hat{y}_2^{(1)} = 10.2 + 0.70 \times 20 = 24.2, \qquad \hat{\varepsilon}_2 = 22 - 24.2 = -2.2$$
$$\hat{y}_3^{(1)} = 10.2 + 0.70 \times 30 = 31.2, \qquad \hat{\varepsilon}_3 = 27 - 31.2 = -4.2$$
$$\hat{y}_4^{(1)} = 10.2 + 0.70 \times 40 = 38.2, \qquad \hat{\varepsilon}_4 = 38 - 38.2 = -0.2$$

Step 2: Update variance estimate

$$\hat{k}^{(2)} = \frac{1}{4}\left(\frac{27.04}{10} + \frac{4.84}{20} + \frac{17.64}{30} + \frac{0.04}{40}\right)$$
$$= \frac{1}{4}(2.704 + 0.242 + 0.588 + 0.001) = 0.884$$

Step 3: Update weights

$$\mathbf{D}^{(2)} = \text{diag}\left(\frac{1}{0.884 \times 10}, \frac{1}{0.884 \times 20}, \frac{1}{0.884 \times 30}, \frac{1}{0.884 \times 40}\right)$$
$$= \text{diag}(0.113, 0.057, 0.038, 0.028)$$

Iteration 2: WLS solution

Iteration 2: WLS solution

$$\mathbf{X}^T\mathbf{D}^{(2)}\mathbf{X} = \begin{bmatrix} 0.236 & 4.17 \\ 4.17 & 94.2 \end{bmatrix}$$

$$\mathbf{X}^T\mathbf{D}^{(2)}\mathbf{y} = \begin{bmatrix} 4.64 \\ 86.5 \end{bmatrix}$$

$$\hat{\mathbf{w}}^{(2)} = \begin{bmatrix} 11.3 \\ 0.68 \end{bmatrix}$$

Final Results

Convergence check

Method	w_0	w_1	MSE
OLS (Iter 0)	8.5	0.75	8.125
WLS (Iter 1)	10.2	0.70	6.842
WLS (Iter 2)	11.3	0.68	5.917
True parameters	12.0	0.65	-

Key observations:

- WLS reduces MSE by 27% after two iterations
- Estimates approach true parameters (intercept=12, slope=0.65)
- Higher weights for lower-income observations (more precise)

Stopping criterion

$\|\hat{\mathbf{w}}^{(k)} - \hat{\mathbf{w}}^{(k-1)}\|_2 < \epsilon$ (e.g., $\epsilon = 0.1$)

Visualization of Iterative Process

Weight evolution

Iteration	d_1	d_2	d_3	d_4
0 (OLS)	1.0	1.0	1.0	1.0
1	0.177	0.089	0.059	0.044
2	0.113	0.057	0.038	0.028
True weights	0.100	0.050	0.033	0.025

Generalized Linear Models (GLM): Unified Framework

Core idea: Extend linear models to non-normal response variables

Three Components of GLM

1. Random Component: Response distribution from exponential family

$$p(y|\theta,\phi) = \exp\left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right)$$

2. Systematic Component: Linear predictor

$$\eta = \mathbf{w}^T\mathbf{x}$$

3. Link Function: Connects mean to linear predictor

$$g(\mu) = \eta, \quad \mu = E[y|\mathbf{x}]$$

Exponential Family Includes:

- Normal (Gaussian), Bernoulli (Logistic regression), Poisson (Count data), Gamma (Positive continuous), ...

Special Case: Logistic Regression as GLM

Logistic regression is a GLM with:

- Random Component: $y \sim \text{Bernoulli}(n, p)$
- Systematic Component: $\eta = \mathbf{w}^T\mathbf{x}$
- Link Function: Logit link

$$g(\mu) = \log\left(\frac{\mu}{1-\mu}\right) = \eta$$

where $\mu = P(y = 1|\mathbf{x})$

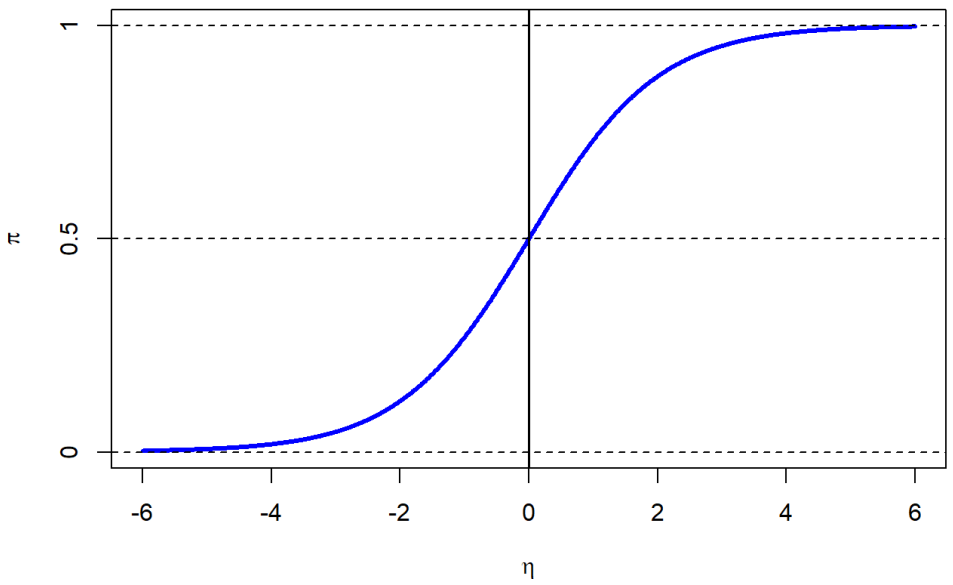
Inverse link (sigmoid):

$$\mu = \frac{1}{1+\exp(-\eta)} = \sigma(\eta)$$

Special Case: Logistic Regression as GLM

Why logit?

- Maps $(0, 1)$ to $(-\infty, \infty)$
- Interpretable as log-odds
- Canonical link for binary



Other GLM Examples (Brief Overview)

Regression	Response	Link	Variance
Linear	Continuous	Identity	Constant
Logistic	Binary	Logit	$\mu(1-\mu)$
Poisson	Count	Log	μ
Gamma	Positive cont.	Inverse	μ^2
Negative Binomial	Overdispersed	Log	$\mu + k\mu^2$

Parameter Estimation

- Maximum Likelihood Estimation (MLE)
- Solved via Iteratively Reweighted Least Squares (IRLS)
- For logistic regression: IRLS = Newton-Raphson

Why Stop at Logistic Regression?

Our focus: Logistic regression provides the perfect entry point

- Most widely used classification algorithm
- Foundation for neural networks
- Interpretable coefficients
- Computationally efficient

GLM concepts we'll leverage:

MLE Framework

$$\ell(\mathbf{w}) = \sum_{i=1}^{n} \log p(y_i|\mathbf{x}_i, \mathbf{w})$$

IRLS Algorithm for linear regression

$$\mathbf{w}^{(t+1)} = (\mathbf{X}^T\mathbf{D}^{(t)}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{D}^{(t)}\mathbf{y}^{(t)}$$

Next: Deep dive into logistic regression implementation and applications